

HPC/Training Doc

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▼ Conda Commands

- To create a new environment: `conda create --name myenv`
- To activate an environment: `conda activate myenv`
- To deactivate an environment: `conda deactivate`
- To list all environments: `conda env list`
- To remove an environment: `conda env remove --name myenv`
- To update Conda to the latest version: `conda update conda`

▼ Slurm Commands

- To submit a job: `sbatch job_script`
- To view the queue: `squeue -u username` / `squ`
- To cancel a job: `scancel job_id`
- To view job information: `scontrol show job job_id`

- To view node information: `scontrol show node node_name`

▼ Steps After Getting a Cluster

- Source profile for Slurm and env vars: `source ~/.bashrc; source /etc/profile`
- Load slurm modules: `module restore recycling_module`
- Load Anaconda: `m1 anaconda`
- Load env: `conda activate recyclingEnv`
- Pull latest changes: `git pull`
- Install new packages: `pip install -r requirements.txt`
- Run training: `python3 train.py --noplots --data ../far_west_7_set/data.yaml --weights yolov5s.pt --batch-size 32 --seed 42 --tracking_uri http://ec2-18-220-114-91.us-east-2.compute.amazonaws.com:5000/`

▼ Creating a Fresh Conda Environment

There has been an issue where a large batch size will cause the program to crash and a new environment is needed for the program to be runnable again.

Getting `Segmentation Fault` when there are no changes, try reinstalling environment

Follow these steps to create and set up a fresh Conda environment:

1. Source profile for Slurm and env vars:

```
source ~/.bashrc; source /etc/profile
```

2. Load anaconda

```
m1 anaconda
```

3. Remove environment

```
conda remove -n recyclingEnv --all
```

4. Create environment

```
conda create --name recyclingEnv --y
```

5. Activate Environment

```
conda activate recyclingEnv
```

6. install requirements

```
conda install --yes --file requirements.txt
```

7. Install pip

```
conda install pip
```

8. pip install remaining packages that are not available on conda

```
pip install -r requirements.txt
```

9. Single-line command:

```
source ~/.bashrc && source /etc/profile && ml anaconda && conda remove -n recyclingEnv --all --y && conda create --name recyclingEnv --y && conda activate recyclingEnv && conda install --yes --file requirements.txt && conda install pip && pip install -r requirements.txt
```

▼ Train.py Errors Encountered

This section will be dedicated to documenting any errors encountered during the execution of `train.py`

- Error 1: Killed at first epoch before batch finishes
lack of memory, check
`-mem` tag on cluster creation
- Error 2: Getting `Segmentation Fault` when there are no changes
try reinstalling environment
- Error 3: mAP plummets after 3-4 epochs
There was an error with importing MLFlow causing memory to crash
- Error 4: Segmentation Fault
There was an error with importing MLFlow causing memory to crash
- Error 5: `Killed` before 1st batch on 1st epoch
There was an error with importing MLFlow causing memory to crash

▼ Using `hpc_run.sh`

The following are tips on how to use the script:
e.g.:

```
./hpc_run.sh train --tracking_uri http://ec2-3-16-26-164.us-east-2.compute.amazonaws.com:5000/ --data /nfs/stak/users/keel/hpc-temp/datasets/testA_1000/far_west_4_set/data.yaml --weights yolov5s.pt --seed 42 --batch-size -1 --epochs 150 --patience 30 --workers 0
```

- `./hpc_run.sh train`

- The HPC auto appends `--no-plots` to the execution of `train.py`
- Tracking run with MLFlow:

```
--tracking_uri http://ec2-3-16-26-164.us-east-2.compute.amazonaws.com:5000/
```

- Using custom dataset:

```
--data /path/to/data.yaml
```

- Using specific weights:

```
--weights path_to_weight.pt
```

```
--weights MLFlow
```

→ Download weights from MLFlow, must set `--tracking_uri` (WIP)

- End training if the model is not improving:

`--patience 30` → ends training after 30 epochs if model accuracy is not improving

- Setting batch size:

```
--batch-size 32
```

- Using AutoBatch; dynamically allocating batch size for full GPU utilization:

```
--batch-size -1 --workers 0
```

- `./hpc_run test_server`

- Runs `test_server.py` with blocking and interactive mode; `srun`
- This is just to test a model with inference
- This is ran with `--hpc` arg to exclude the `keyboard` module

- `./hpc_run clear`

- Deletes local output folder
- Deletes logs on the hpc
- Deletes saved experiments on the HPC repo
- Does **NOT** delete runs logged to MLFlow

▼ Information about OSU's HPC cluster:

- HPC setup guide:
 - <https://it.engineering.oregonstate.edu/hpc>
- About:
 - <https://it.engineering.oregonstate.edu/hpc/about-cluster>
- HPC Manager:
 - Robert Yelle
 - robert.yelle@oregonstate.edu

▼ AWS `hpc_run.sh` script (copy)

```
#!/bin/bash

MLFlow_tracking_uri="http://ec2-3-16-26-164.us-east-2.compute.amazonaws.com:5000"
repo_dir=~/.hpc-share/ai-recycling/ai-recycling"

if [ "$1" = "test_server" ]; then
    shift
    ssh submit -t "bash -l -c \"cd $repo_dir && ./srun_
elif [ "$1" = "train" ]; then
    shift
    ssh submit -t "bash -l -c \"cd $repo_dir && ./sbatch
elif [ "$1" = "clear" ]; then
    rm -rf outputs/*
    ssh submit "rm -rf $repo_dir/../../log_srun.log $repo_
else
    echo "Invalid argument used: \"$1\""
```

```
        echo "try: ssh_hpc [\"test_server\"/\\"train\"/\\"cle  
fi
```

Other Documentations